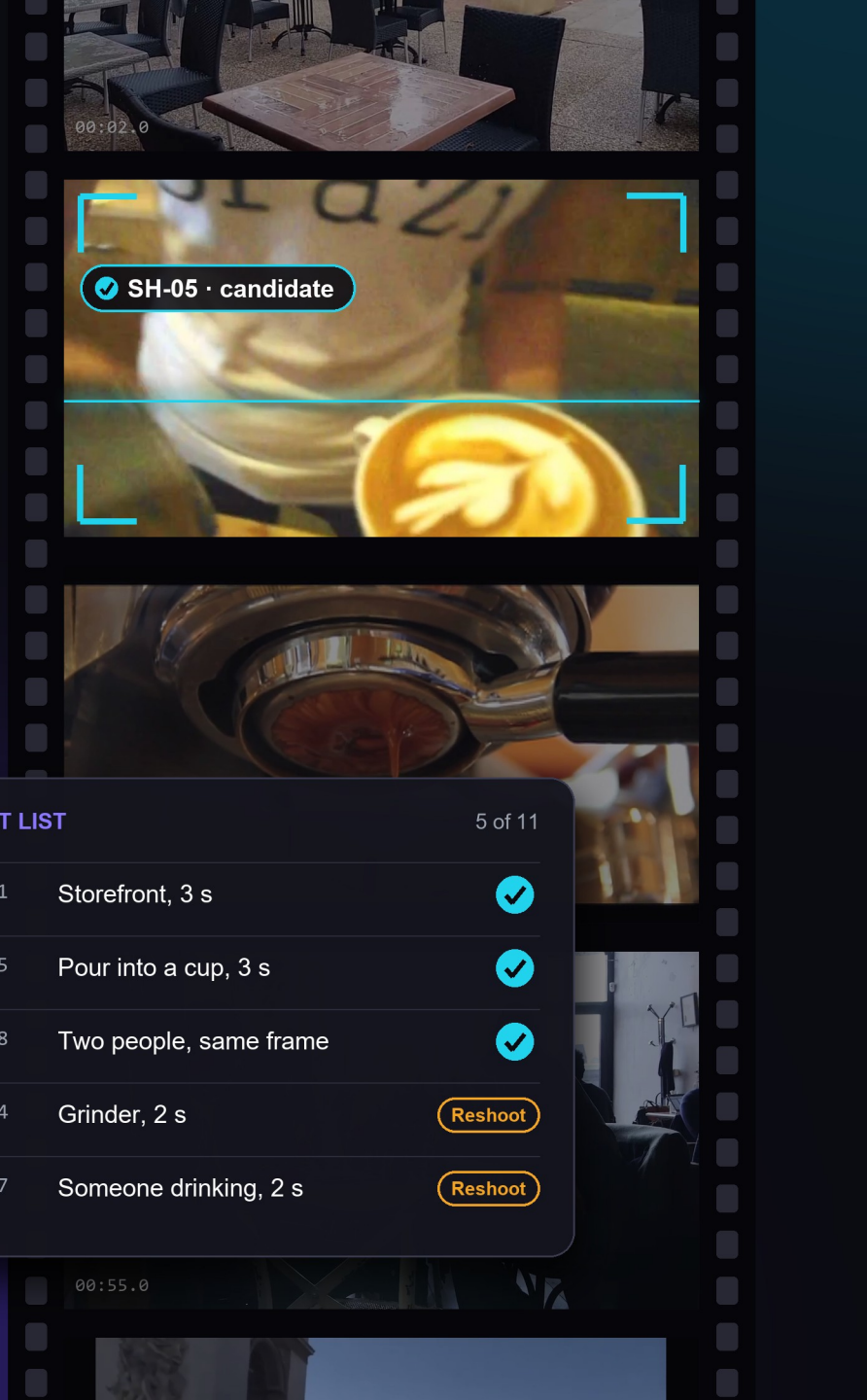
CREATOR AI · LIGHTNING TRACK · ASUS UGEN300

# SceneGap Edge

Helps a crew check the shot list against the footage before leaving the location, with playable evidence and a person's decision on every shot.

- Runs on your own computer
- Evidence for every answer
- A person decides
- UGen300 port planned




Stage I proposal · [Team name] · github.com/[team]/scenegap-edge

PROBLEM

# The missing shot is found at 23:00, after the café has closed



**Illustrative scenario · two-person crew, café promo, 11 must-have shots**

*“We had pours. None of them was three seconds long. We found out in the edit.”*

| | |
|---|---|
| **09:00** | Shot list agreed with the client: storefront, interior, beans, grinder, 3 s pour, two barista takes, bag with readable brand. |
| **17:40** | Wrap. Several cards of footage; nobody has time to watch it all on site. |
| **23:00** | The editor scrubs for the pour: every take cuts away after 2 s, and nobody filmed a customer drinking. |
| **Next week** | Reshoot day: NT$12,000–28,000 per crew-day in Taiwan, US$800–1,500 per videographer-day in the US, plus café and client again. |

Day rates: PRO360 quote statistics (TW), Film Lifestyle mid-level rates (US); docs/research/BUSINESS_FACTS.md.

PROBLEM

# Existing tools find footage; SceneGap adds shot-list rules, evidence and reshoot tracking

| TOOL | FINDS THE MOMENT IN A CLIP | SHOT-LIST RULES (LENGTH, TAKES, SAME FRAME) | RUNS LOCALLY |
|---|---|---|---|
| **Premiere Pro media intelligence** | yes, In/Out range | not in official docs | yes, on device |
| **DaVinci Resolve 21 IntelliSearch** | visual search: whole clips | not in official docs | not stated |
| **Final Cut Pro 12 visual search** | yes | not in official docs | not stated |
| **invideo AI script supervisor** | not stated | done / pending only | no (upload) |
| **SceneGap Edge** | **yes, original timecode** | **checked per rule; a person decides** | **yes** |

> **The difference is the workflow around search: shot list → rules → evidence → a person's decision → reshoot list → re-check.**

Vendor documentation, accessed 2026-10-26 (Adobe media intelligence FAQ and others). More tools and prices: A11.

PROBLEM

# What works today, and what still needs proof

## Shown today (live app, real footage)

- Shot list in the crew's words, with rules: length, takes, same frame
- Candidate clips with playable evidence on the original timecode
- Rule checks: 10 of 12 controlled cases judged right (slide 12)
- A person accepts, rejects, or adds the shot to the reshoot list
- Demo: 9 Commons clips (5.9 min) analysed in 22 s on a desktop GPU; new footage re-checked file by file

## Not shown yet

- Automatic missing-shot detection: 0 of 38 caught on test (slide 15)
- Time saved: no user study yet. Proxy: 317 s of candidate clips to play vs 839 s of footage (38%)
- UGen300 speed and quality: not run on the device yet
- Human-checked labels: current labels are AI-assisted

**SceneGap Edge today is a review assistant that keeps evidence and a person's decision; automation grows only as it is proven.**

SOLUTION

# From your shot list to a decision you can play back

### 1 · Write the shot list

In Chinese, as the crew speaks; a person checks the English search phrase and adds rules. Test list: 11 shots, each with a minimum length; 1 same-frame, 1 two-takes, 1 manual brand check.

### 2 · Index footage locally

Camera cuts, one frame per second on the original timecode, CLIP vectors. Desktop: 2.3 s per minute of video (GPU), 3.1 s (CPU). UGen300 next.

### 3 · Check each rule

Pass, fail or can't tell per rule; failing clips stay visible. Threshold τ 0.225 set on dev; search over 10,000 segments p95 14 ms.

### 4 · A person decides

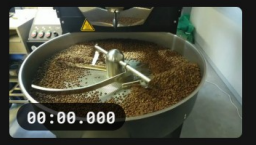
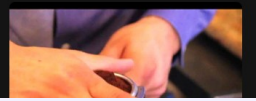
Accept, reject with a reason, or add to the reshoot list. Only the person's decision counts as done; it is bound to the shot's version.

## The AI's four answers (a person's decision is stored separately)

| AI ANSWER | MEANING | ON THE 26-CLIP TEST LIBRARY |
|---|---|---|
| **Candidate** | Every rule passes on at least one clip; still needs a person to accept | 10 of 11 shots |
| **Partial** | Something matched but a rule failed or can't be decided (too short, one take of two, a manual check) | 1 (SH-10, manual brand check) |
| **Not found** | Nothing reached the threshold in the analysed files; not proof of absence | 0; 30 of 38 missing shots still got 'candidate' (slide 15) |
| **Unverifiable** | Files failed to decode or are missing, so absence cannot be claimed | 0 |

SOLUTION

# Before leaving, the crew sees which shots are confirmed

| ID | 鏡頭 | AI 判斷 | 人工決定 | 原因 | 片段 | |
|---|---|---|---|---|---|---|
| SH-01 v1 | **店外街景，看得到咖啡店門面，連續 3 秒**<br>a photo of the outside of a cafe seen from the street | AI ◎ 有候選 | ✓ 已接受 | – |  <br>已接受 · 00:00.000–00:10.012 | 看片段 |
| SH-02 v1 | **店內全景，看得到桌椅，連續 3 秒**<br>a photo of the inside of a cafe with tables and chairs | AI ◎ 有候選 | ✓ 已接受 | – |  <br>已接受 · 00:55.013–00:59.017 | 看片段 |
| SH-03 v1 | **咖啡豆特寫，連續 2 秒**<br>a close-up photo of roasted coffee beans | AI ◎ 有候選 | ○ 未確認 | – |  <br>00:00.000–00:09.000 | 看片段 |
| SH-04 v1 | **磨豆過程，連續 2 秒**<br>a photo of a coffee grinder | AI ◎ 有候選 | ○ 未確認 | – |  | 看片段 |

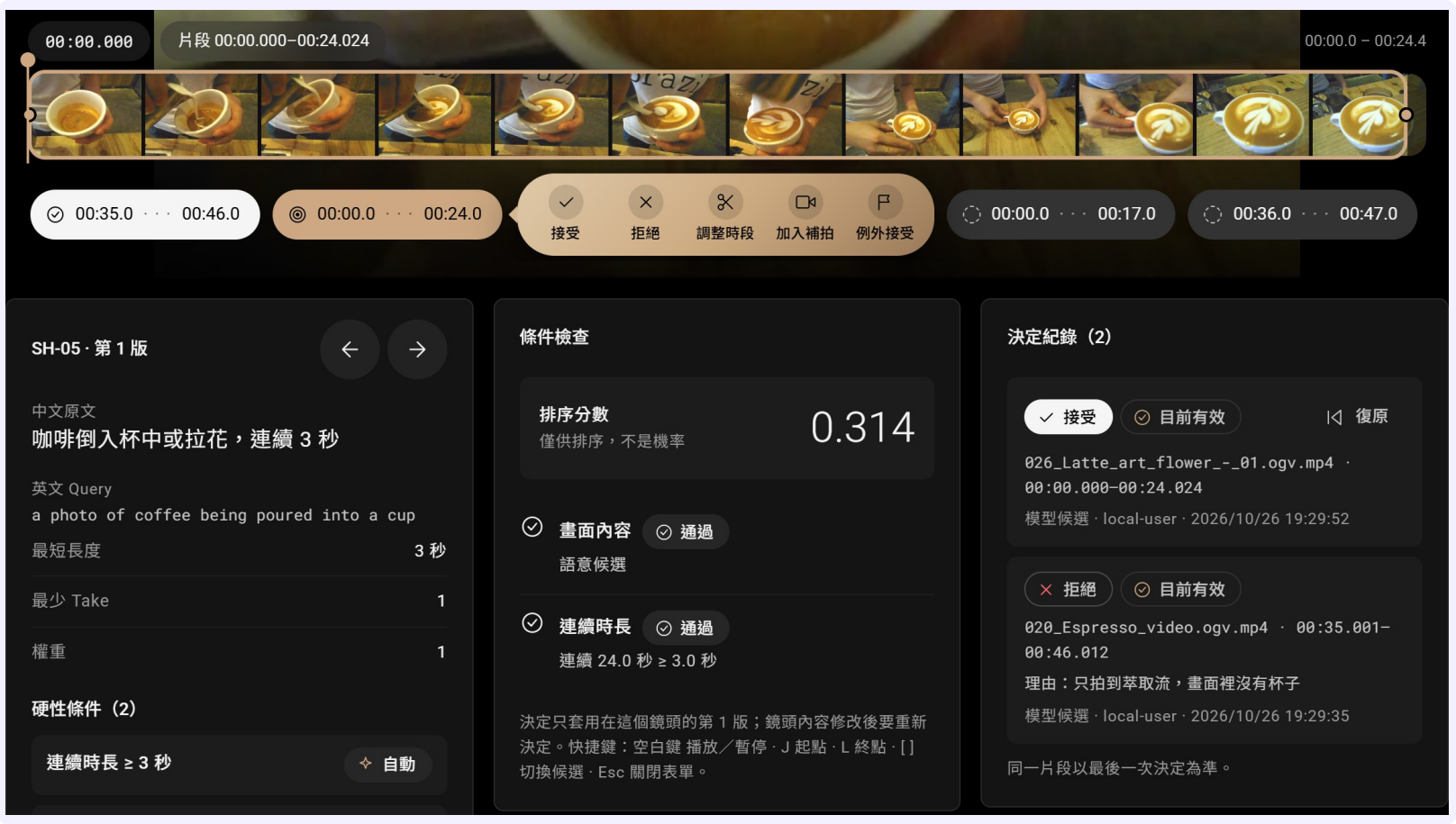
## This demo project

- 9 Commons clips, 5.9 min, analysed in 22 s on a desktop GPU
- AI candidate: 10 of 11 shots
- Accepted by a person: 5 of 11 (SH-11 needs a second take)
- Partial: SH-10 (manual brand check)
- Reshoot list: grinder, drinking, coffee bag
- Decisions entered as a scripted demo review, not timed

Live app (Coverage page, cropped). Columns: shot · AI answer · person's decision · reason · clip.

SOLUTION

# Every answer plays its clip, and a person decides




## SH-05 on this screen

- Shot 「咖啡倒入杯中或拉花，連續 3 秒」 as committed before any frame was seen (eval/requirements.json, 10/26 18:09)
- Filmstrip and clip ranges on the original timecode
- Ranking score 0.314: orders clips, not a probability
- Length check 24.0 s ≥ 3 s: how long frames stay close to the query
- Log: espresso stream with no cup rejected with a reason; latte-art pour accepted; each can be undone

Live app (clip view, cropped). Original files are never touched.

SOLUTION

# Shots without a usable clip go on the reshoot list



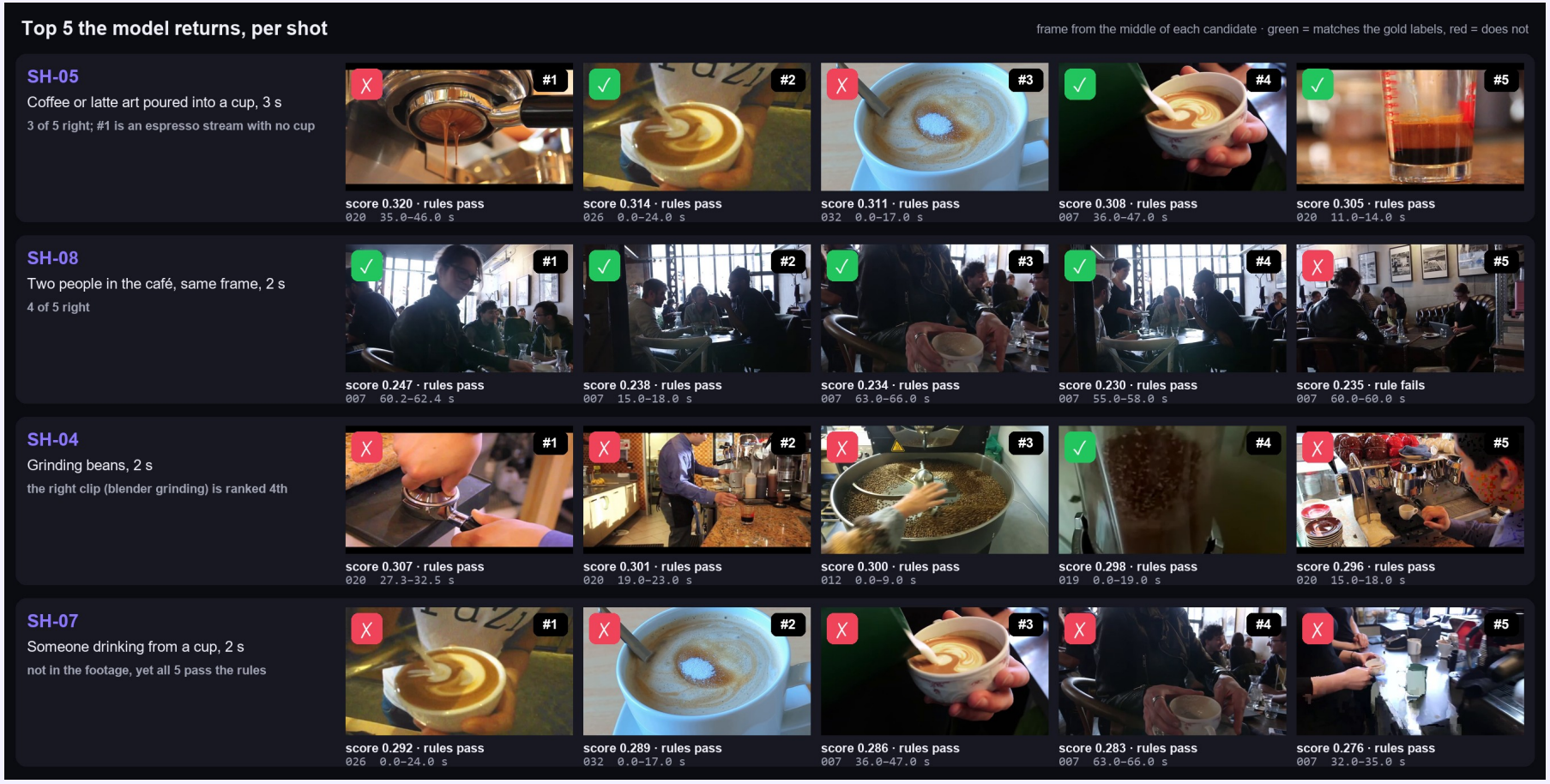

## What the crew gets

- One card per shot: why it is missing, the closest clip, minimum length, how it will be checked
- The card shows the AI's rules passed; the person decided the clip shows tamping, not grinding
- Status: to decide → reshoot → shooting → done / not reshooting
- Video demo: the reshot clip was analysed alone in 2.5 s and the grinder shot left the list
- The demo review stopped at the top 3; the 4th candidate was a valid grinding clip (next page)
- Fixed: the card now shows how many candidates a person checked ( 已檢查 3 ／ 5) and warns when some are unchecked

Live app (reshoot list, SH-04 card). Export: CSV / HTML.

EVIDENCE

# What the model returns: top 5 per shot, checked against labels




## Read it as

- SH-05: right clips are in the list, but #1 is wrong
- SH-08: same-frame rule works here (4 of 5)
- SH-04: the right clip is 4th; ranking by peak is the fix
- SH-07: nothing to find, yet 5 clips pass: why missing shots go undetected

Demo project (9 clips); candidates and scores from the live app; right / wrong from eval/labels/gold.json (≥ 0.5 s overlap). decks/scenegap/fig_top5.py.

EVIDENCE

# How the length rule reads the model's scores




## What the curves show

- Street frames score about 0.18; the pour about 0.33, well above τ 0.225
- SceneGap samples 1 frame per second: a 4 s pour gives 4 samples, a lower bound of 3.0 s, so it passes the 3 s rule
- A 2 s pour gives 1.0 s: too short, correctly
- A finished latte scores 0.30 with nothing poured: CLIP sees the result, not the action, so a person checks it
- Stage II: denser sampling near the edges and an action check for pour-type shots

Rule-check cases cut from real Commons footage (eval/rule_check.py); similarity from the same CLIP model at 4 frames/s. decks/scenegap/fig_curves.py.

EVIDENCE

# What each rule actually measures, and what it cannot tell

| RULE | WHAT SCENEGAP MEASURES | WHAT IT CANNOT TELL | TODAY |
|---|---|---|---|
| **Length** | How long frames stay close to the query inside one camera cut (lower and upper bound; re-check every 0.25 s) | Whether the action itself lasts that long; the lower bound runs about 1 s short (4 s pour → 3.0 s) | auto, a person checks |
| **Same frame** | Both phrases above the threshold in the same sampled frame | Who is with whom; objects too small for CLIP | auto, a person checks |
| **Takes** | Distinct files, by content hash; duplicates of one file are one take | A re-encoded copy of the same take | auto, a person checks |
| **Readable brand** | Nothing automatic | Reading text | manual |
| **Not found** | Nothing reached the threshold in the analysed files | Proof the shot was never filmed; failed files make it 'can't tell' | auto, scope shown |

> **Dense sampling gives more observations, not certainty: a clip that looks like pouring is still checked by a person.**

EVIDENCE

# Rule check: 10 of 12 right; both misses are false acceptances

| CASE (REAL COMMONS FOOTAGE, CUT TO A KNOWN ANSWER) | SHOULD | SCENEGAP | MEASURED RUN (S) | FILES / TAKES | RIGHT? |
|---|---|---|---|---|---|
| **SH-05 · continuous 4 s latte-art pour** | pass | pass | 3.0 | 1 / 1 | yes |
| **SH-05 · 6 s pour from a different shoot** | pass | pass | 5.0 | 1 / 1 | yes |
| **SH-05 · pour lasts only 2 s (rule: 3 s)** | not pass | not pass (duration_short) | 1.0 | 1 / 0 | yes |
| **SH-05 · two 2 s pours split by another shot** | not pass | not pass (duration_short) | 1.0 | 1 / 0 | yes |
| **SH-05 · finished latte, nothing is poured (5 s)** | not pass | pass | 4.0 | 1 / 1 | NO |
| **SH-08 · two people at a café table, same frame** | pass | pass | 3.0 | 1 / 1 | yes |
| **SH-08 · two people, but outdoors (not in a café)** | not pass | not pass (no_candidate) | — | 1 / 0 | yes |
| **SH-08 · two people, then a café interior: separate shots** | not pass | not pass (no_candidate) | — | 1 / 0 | yes |
| **SH-11 · one barista clip (rule: 2 takes)** | not pass | not pass (takes_short) | 4.0 | 1 / 1 | yes |
| **SH-11 · the same clip copied byte for byte** | not pass | not pass (takes_short) | 4.0 | 2 / 1 | yes |
| **SH-11 · the same clip re-encoded (still one take)** | not pass | pass | 4.0 | 2 / 2 | NO |
| **SH-11 · two different barista clips** | pass | pass | 4.0 | 2 / 2 | yes |

Open: an action check (a finished latte passes as 'pouring') and re-encoded-take dedup. Such passes stay 'candidate, a person confirms', never done. eval/rule_check.py, frozen threshold.

EVIDENCE

# How we tested: fixed protocol, open footage, AI-assisted labels

Protocol and the 11 shots committed before any label; threshold set on dev only; test scored once. Every file below is in the repo.

## Footage

| ITEM | VALUE |
|---|---|
| **Source** | 44 Wikimedia Commons clips, 25.2 min, H.264 ≤ 1280 px |
| **Dev** | 18 clips, 11.2 min: threshold only |
| **Test** | 26 clips, 14.0 min: scored once |
| **Distractors** | 17 of 44 clips contain none of the 11 shots |
| **Licences** | CC BY-SA 19, CC BY 14, CC0 8, public domain 3 |
| **Split note** | one author pair crosses (022 dev, 025 test); without the 2 projects holding 025, SceneGap still catches 0/31 |

## Labels

| ITEM | VALUE |
|---|---|
| **Labellers** | two independent AI sessions on 1 fps sheets; an AI third pass |
| **Agreement** | 472/484 clip × shot pairs (97.5%); κ 0.914 over 16,687 frame samples |
| **Third pass** | 143 disputed frame labels decided with 9 written rules |
| **Consistency pass** | 78 of 238 agreed positives removed after labelling (65 were SH-11) |
| **Lenient labels** | restoring them leaves SceneGap unchanged (7/11, 18/55) |
| **Threshold** | τ 0.225: dev F1 0.240, frame precision 14.4%, recall 71.8% |

> **Simulated projects reuse the same clips, so they are not independent shoots. Human-checked labels on the original video come next; details in A5–A7.**

EVIDENCE

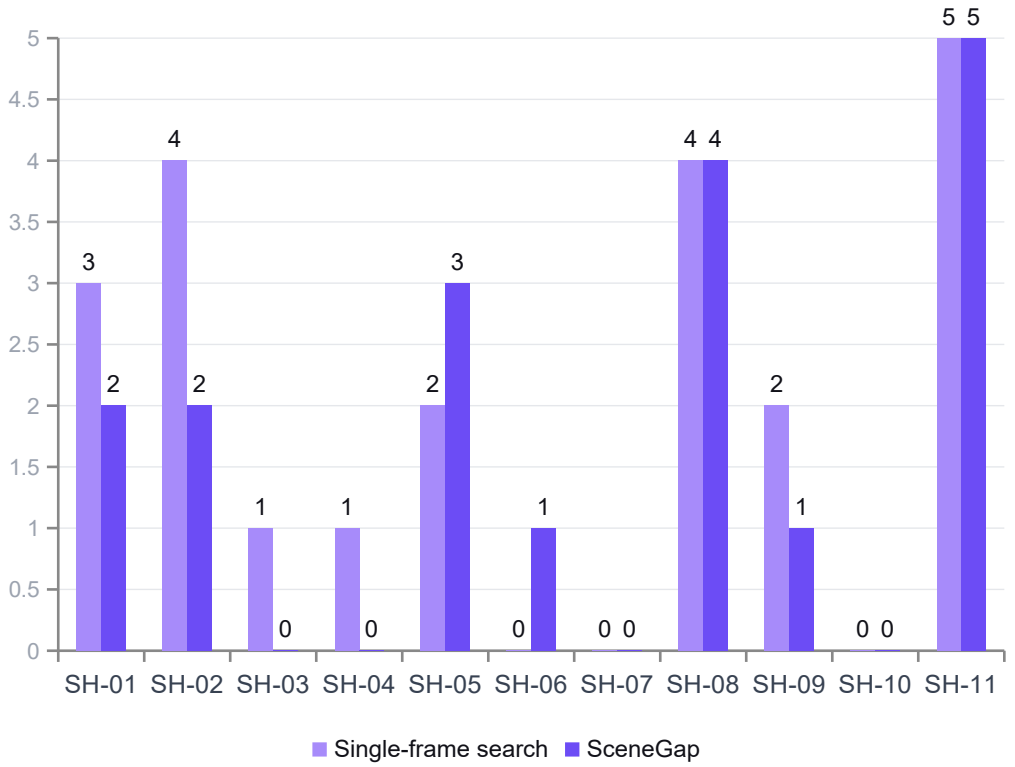
# Right clip in the top 5 for 7 of 11 shots; plain frame search found 8

| METHOD | RIGHT CLIP IN TOP 5 | TOP-5 PICKS RIGHT |
|---|---|---|
| **Title keywords (whole clips)** | 45% (5/11, CI 21–72%) | 36% (8/22, < 5 returned) |
| **Single-frame search** | 73% (8/11, CI 43–90%) | 40% (22/55) |
| **SceneGap, ranked after rule checks** | **64% (7/11, CI 35–85%)** | **33% (18/55)** |

## Why one hit was lost

- Lost SH-03 and SH-04, gained SH-06.
- Clip 019 is the right answer for both. On its own it passes every rule (5 s and 19 s runs): the rules did not reject it.
- Runs are ranked by average similarity; single-frame search uses the peak frame. Stage II ranks by peak.
- A hit needs 0.5 s overlap, so this layer does not test the length rule (slide 12 does).

## Correct clips in the top 5, per shot (of 5)




26 held-out clips, 14.0 min, scored once at τ 0.225; Wilson 95% intervals. SH-03 and SH-07 each rest on one 3 s gold clip. Per shot: A5.

EVIDENCE

# Finding missing shots automatically does not work yet

Unit: one shot in one simulated 8-clip project (8 × 11 = 88): 38 missing, 50 present. Found = 'candidate'; can't decide = 'partial' or 'unverifiable'.

## Test split

| METHOD | MISSING CAUGHT | RESHOOT ASKED, SHOT THERE | CAN'T DECIDE | 'FOUND' WHEN ABSENT |
|---|---|---|---|---|
| **Title keywords** | 34/38 (89%) | 22/50 (44%) | 0/88 (0.0%) | 4/38 (11%) |
| **Single-frame** | 0/38 (0%) | 0/50 (0%) | 0/88 (0.0%) | 38/38 (100%) |
| **SceneGap** | **0/38 (0%)** | **0/50 (0%)** | **11/88 (12.5%)** | **30/38 (79%)** |

### Where the missing shots sit

- SH-07 is missing in 8 of 8 projects; SH-03, SH-04, SH-10 in 6, 6, 6.
- SceneGap answered 'candidate' for 30 of them and 'partial' for 8 (SH-10's manual brand check).
- Why: at τ 0.225 only 14% of frames above the threshold show the shot, so almost every shot finds something.

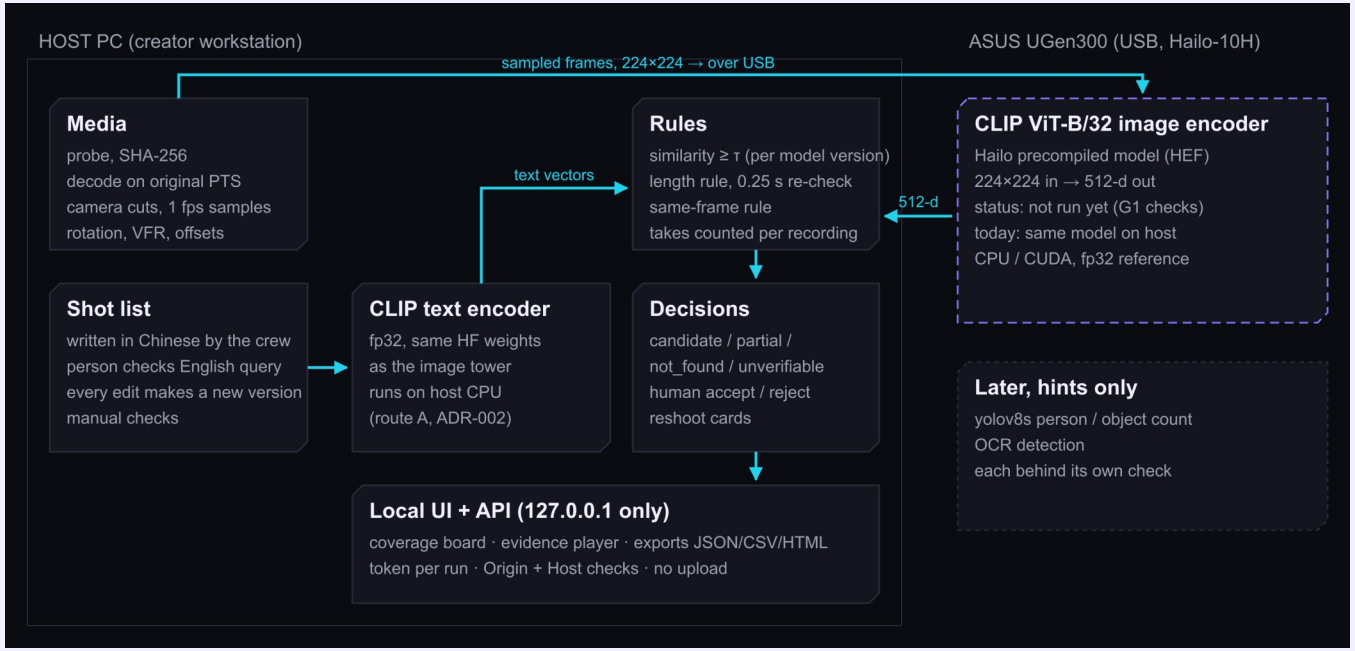
## Dev only, after the test (not used to rescore)

| SCORE | MISSING CAUGHT | WRONG RESHOOT | 'FOUND' WHEN ABSENT |
|---|---|---|---|
| **raw τ 0.225 (frozen)** | 5/89 (6%) | 0/131 (0%) | 80/89 (90%) |
| **raw τ 0.27** | 24/89 (27%) | 14/131 (11%) | 65/89 (73%) |
| **raw τ 0.29** | 41/89 (46%) | 40/131 (31%) | 37/89 (42%) |
| **margin 0.02** | 16/89 (18%) | 7/131 (5%) | 66/89 (74%) |
| **margin 0.0475** | 29/89 (33%) | 14/131 (11%) | 51/89 (57%) |

**Target (PRD 21.5): wrong reshoot ≤ 5%. Inside it on dev: margin 0.0075, 11/89 caught at 0/131. Margin 0.02 gives 16/89 at 7/131 = 5.3%, just over.**

HARDWARE

# Plan: the image half of CLIP on the UGen300; the PC keeps the rest




## ASUS UGen300

| ITEM | VALUE |
|---|---|
| **Chip** | Hailo-10H |
| **AI performance** | 40 TOPS INT4 / 20 TOPS INT8 |
| **Memory** | 8 GB LPDDR4, 4266 MT/s |
| **Link** | USB 3.1 Gen2 Type-C, 10 Gbps |
| **Power / size** | 2.5 W typical; 105 × 50 × 18 mm, 150 g |
| **OS** | Windows, Linux, Android (no macOS listed) |
| **Price** | $299.99 USB 8 GB (reported) |
| **CLIP ViT-B/32 image encoder** | 81.5 / 350 FPS at batch 1 / 8 (Hailo zoo, PCIe) |

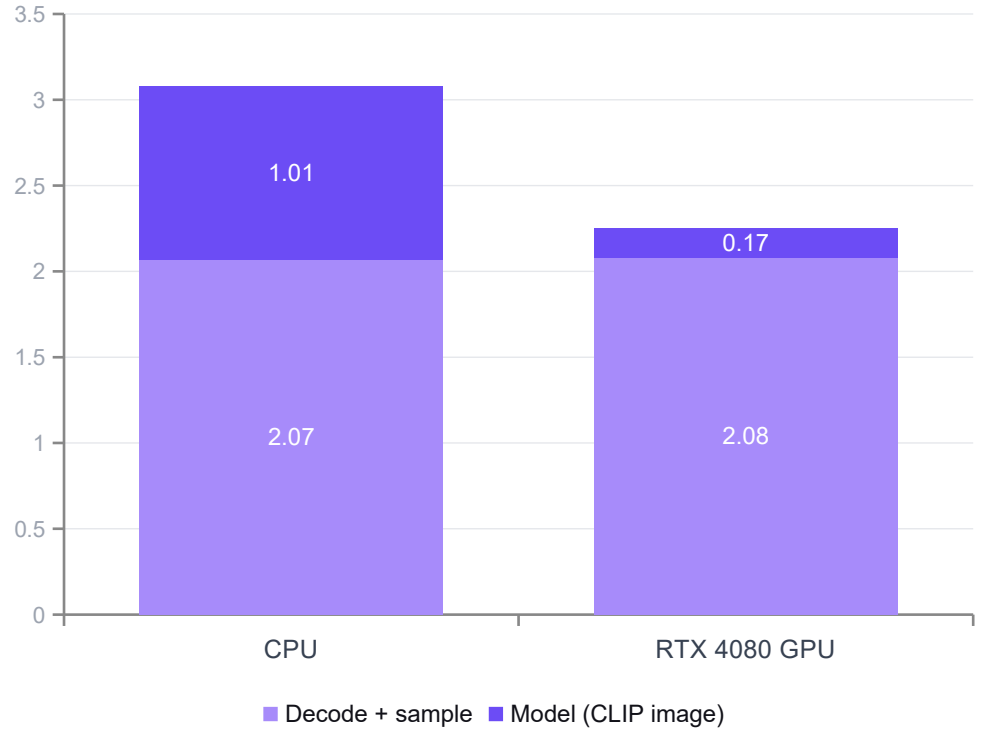
Frames over USB: about 9 MB per minute of video (estimate) on a 10 Gbps link. Before the device replaces the PC model it must pass six G1 checks against the PC (A1). Demo PC: Windows 11, RTX 4080; on a Mac SceneGap runs on CPU only.

BUSINESS_FACTS.md §2 (ASUS spec; price reported by CNX Software), MODEL_PLAN.md §2, §6; Hailo Model Zoo measures over PCIe, not USB.

HARDWARE

# Measured: decoding is 2.1 of the 2.3 s per minute; a UGen300 would free the GPU or a laptop CPU

## Seconds per minute of video (desktop)




## Model and search

| BACKEND | MODEL SPEED | MODEL LOAD | SEARCH P50 / P95 (10,000 SEGMENTS) |
|---|---|---|---|
| **CPU** | 61.5 frames/s | 6.51 s | 11.77 / 14.29 ms |
| **RTX 4080** | 365 frames/s | 1.88 s | 7.61 / 13.39 ms |
| **UGen300 (vendor)** | 81.5 / 350 FPS (batch 1 / 8, PCIe) | not run | not run |

## Playing a 1080p30 test pattern while analysing (budget 33.3 ms)

| ANALYSIS | LATE FRAMES | FRAME WORK P95 | ANALYSES FINISHED |
|---|---|---|---|
| **none** | 0 / 1,800 | 5.7 ms | 0 |
| **CPU** | 0 / 1,800 | 7.5 ms | 22 |
| **GPU** | 0 / 1,800 | 6.2 ms | 52 |

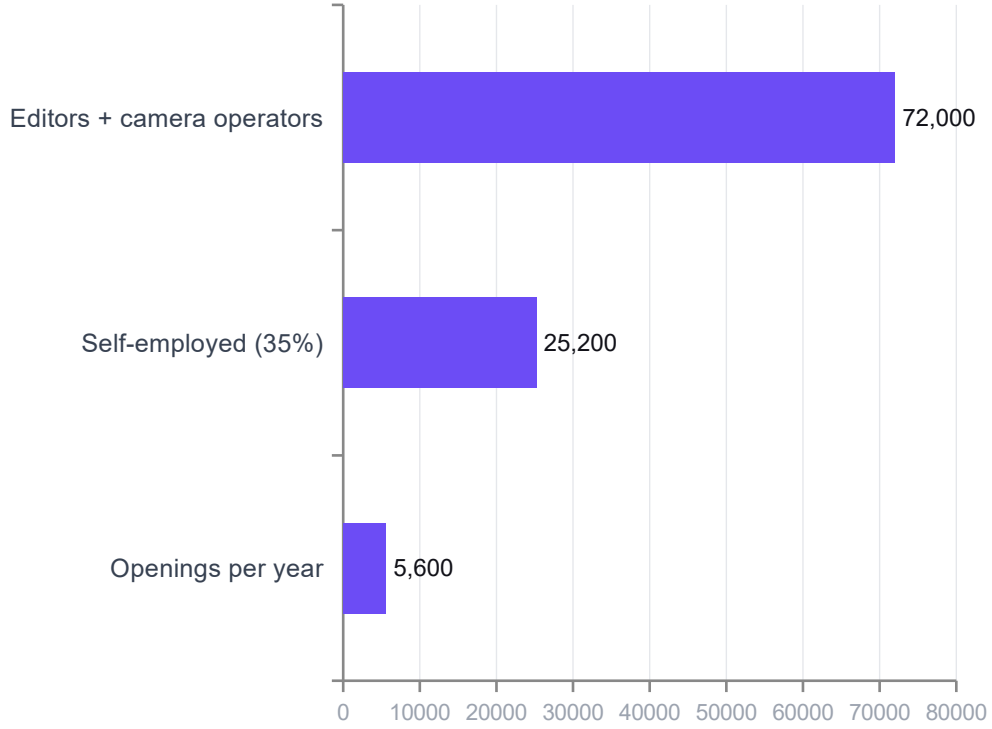
> **The model is 32% of the CPU run and 7% of the GPU run. Stage II measures an editing app alone vs with CPU, GPU or UGen300 analysis.**

eval/results/perf.json (six short clips, 85 s, ≤ 1280 px, warm model; import and hashing not timed), interference.json (synthetic test pattern, not an editing app).

BUSINESS

# Sourced: who pays first is 1–3 person crews and freelance editors

## US floor (BLS OOH, 2025)




## Segments (PRD 4.1)

| SEGMENT | SIZE SIGNAL | MUST HOLD TRUE |
|---|---|---|
| **1–3 person crews (first)** | ≈ 11,400 self-employed US camera operators; TW crew-day NT$12,000 –28,000 | they write a shot list and carry a computer on site |
| **Freelance editors (first)** | ≈ 13,800 self-employed US editors; median $75,420/yr | client briefs are specific; a pickup can still be arranged |
| **Student teams** | no data | will check a little by hand |
| **Brand content producers** | 91% of businesses use video (Wyzowl 2026, partial) | templates stay stable |

BLS is a floor: it leaves out YouTubers and in-house marketing staff. Worldwide, ~50M creators and ~4% professional (Goldman Sachs, partly verified).

US BLS OOH accessed 2026-10-26 (39,400 editors + 32,600 camera operators; 35% self-employed; ~5,600 openings a year); self-employed split is our arithmetic. Day rates per videographer.

BUSINESS

# Hypothesis: one prevented reshoot could offset the device cost

## Value = hours saved × hourly cost + reshoot avoided − tool cost

| TERM | VALUE USED | STATUS |
|---|---|---|
| **Hourly cost, editor** | $36.26/h = $75,420 ÷ 2,080 h | sourced; 2,080 h assumed |
| **Reshoot day, US, 2 people** | $1,600–3,000 = 2 × $800–1,500 | sourced rates; crew of 2 assumed |
| **Reshoot day, Taiwan** | NT$12,000–28,000 per crew-day | quote statistics |
| **Tool cost** | $299.99 UGen300; core software free | price reported |
| **Hours saved** | unmeasured (proxy: 38% of footage to play) | user study, Stage II |
| **Reshoots avoided per project** | unmeasured | interviews, Stage II |

**8.3 h**
of editor time pays for the device

**0.10–0.19**
of one US two-person reshoot day

## Price per seat, USD per year or one-time




Open core: the Apache-2.0 checker is free. Paid: a tested UGen300 bundle (installer, driver versions, device-checked model, support) and shot-list templates. Axle FOCUS: $2,995+/yr.

**Device price only: the on-site computer, setup, upkeep and the person's time are extra. Test it: two groups check one card, by hand vs with SceneGap.**

BUSINESS_FACTS.md §1c, §2, §4; * vendor page not readable at access. Yearly = 12 × monthly list price.

PLAN

# Shown today, proven next

SceneGap Edge today: shot list → candidate clips with evidence → a person's decision → reshoot list → re-check, on real footage.

| PROOF | TODAY | STAGE II TARGET | SOURCE |
|---|---|---|---|
| **1 · Rules** | 10 of 12 controlled cases; right clip in top 5 for 7/11; missing shots caught 0/38 | Recall@5 ≥ 90% and wrong reshoot ≤ 5% in the supported scope; ranking by peak; per-shot thresholds | PRD 21.5 |
| **2 · Labels** | AI-assisted, κ 0.91 | a person labels sampled pairs on the original video | Protocol |
| **3 · People** | no study; proxy 38% of footage to play | 3–5 interviews, then 8–12 crews; median task time −30% | PRD 22.1 |
| **4 · UGen300** | not run; PC harness cos 1.0000, ρ 0.99999 | G1a cos ≥ 0.98, G1b ρ ≥ 0.95, G1e status ≥ 90%; USB speed, power, editing-app load | MODEL_PLAN §4 |
| **5 · Speed** | search p95 13–14 ms; 2.3 s per minute (GPU) | first result for a 60 s clip ≤ 45 s, timed end to end | PRD NFR-05 |
| **6 · Software** | 47 automated tests; 30 of 36 acceptance cases pass; clean rebuild passes | the device and user cases (TC-017, 018, 033, 034) | A9, A12 |

**Fewer surprises in the edit: a reshoot list made before the crew leaves, with the evidence a person checked.**

APPENDIX

# A1 · G1: six checks before the UGen300 replaces the PC model

| CHECK | MEASURE (DEV FRAMES, SAME PREPROCESSING) | PASS BAR | STATUS |
|---|---|---|---|
| **G1a** | Cosine, HEF vs fp32 embedding, per frame | median ≥ 0.98, p5 ≥ 0.95 | not run yet |
| **G1b** | Spearman ρ of similarity per query | median ≥ 0.95, worst ≥ 0.85 | not run yet |
| **G1c** | Top-5 / top-10 shot overlap | ≥ 0.80 / ≥ 0.85 | not run yet |
| **G1d** | Dev Hit@5, each backend with its own τ | no dev shot lost vs fp32 | not run yet |
| **G1e** | Status agreement on the dev project simulation | ≥ 90% same status; no new 'candidate' on a true gap | not run yet |
| **G1f** | Frames/s over USB, 60 s clip end to end | reported, no bar | not run yet |
| **Harness** | Same script, CPU vs CUDA fp32 (662 frames, 13 queries; plan ≥ 2,000 / ≥ 40) | ≈ 1.0 expected | cos 1.0000, ρ 0.99999 |

**Vendor: Hailo-10H ViT-B/32 image encoder, CIFAR-100 zero-shot 65.3 float → 64.5 on chip (PCIe, not USB).**

docs/MODEL_PLAN.md §2, §4; eval/results/g1_selfcheck.json; Hailo Model Zoo (vendor figures).

APPENDIX

# A1b · After G1: product-level checks on the device

| CHECK | WHAT IT PROTECTS AGAINST | STATUS |
|---|---|---|
| **Status agreement, UGen300 vs PC, on dev projects** | quantisation causing more false acceptances | not run yet |
| **USB end-to-end analysis time for a 60 s clip** | a device that is fast in the zoo but slow over USB | not run yet |
| **Editing-app playback and scrubbing with analysis on the device** | claiming host relief without measuring it | not run yet |
| **Whole-system extra power (meter at the wall)** | quoting 2.5 W as the system's cost | not run yet |
| **Missing shots caught and wrong reshoots on new footage, verifier on the device** | the product effect, not just matching vectors | not run yet |

> **Model agreement (G1) is necessary, not sufficient: the product effect is measured on new footage.**

APPENDIX

# A2 · What could break, and what we do then

| RISK | FALLBACK |
|---|---|
| **HEF fails G1 (quantisation hurts ranking)** | Recalibrate with dev frames; TinyCLIP; else ship the CPU/GPU version and say so |
| **A borderline camera cut splits a long take** | The length check says 'can't tell', not 'too short'; the cut is shown |
| **One global threshold fits some shots badly** | One threshold per model version now; per type of shot next, set on dev |
| **English phrase drifts from the Chinese shot** | A person checks the phrase; a multilingual model only after its own check |
| **AI-assisted labels, Commons footage, small test** | Intervals shown; Stage II adds human labels and self-shot projects |
| **Crews will not write shot lists** | Templates; the interviews check this first |

APPENDIX

# A3 · Stage II order: rules and people first, then the device

| Wk 1–2 | Wk 2–3 | Wk 3–5 | Wk 4–6 | Wk 6+ |
|---|---|---|---|---|
| **Rules and labels** | **Missing shots** | **User study** | **UGen300** | **Chinese queries** |
| More controlled cases; a person labels sampled pairs on the original video; rank runs by peak. | Per-shot thresholds with a project-level target; preregister, score once on new footage. | 3–5 interviews, then 8–12 crews: by hand vs SceneGap, counterbalanced; time, misses, wrong reshoots. | G1 on the device; editing-app load with CPU, GPU or UGen300 analysis. | A multilingual model (SigLIP 2), adopted only if it finds as many dev shots. |

APPENDIX

# A4 · Footage shown in this deck

| # | WIKIMEDIA COMMONS FILE | AUTHOR | LICENCE |
|---|---|---|---|
| 001 | Vichy - Grand Café, terrasse sous la grêle.ogv | Romainbehar | CC0 |
| 007 | Café Lomi, Paris - Vimeo.webm | ideafarm films | CC BY 3.0 |
| 012 | Kaffe - Ystad-2024.ogg | Jonn Leffmann | CC BY 3.0 |
| 016 | Lions Almelo zamelt koffie in voor Voedselbank.webm | AAVISIE | CC BY 3.0 |
| 019 | MoliendoCafeEnLicuadora ohs02.webm | PetrohsW | CC BY-SA 4.0 |
| 020 | Espresso video.ogv | Rosatrieu | CC BY-SA 3.0 |
| 026 | Latte art flower - 01.ogv | Anna Frodesiak | CC0 |
| 032 | Sugar falling into coffee.webm | Anselm Schüler | CC BY-SA 4.0 |
| 041 | Empty Streets of Paris COVID19.webm | CreatorsCloud | CC BY 3.0 |

**Screenshots and cover frames come from these clips. All 44 clips used in the evaluation, with links: docs/CREDITS.md.**

APPENDIX

# A5 · The 11 shots and per-shot results (correct / returned, top 5)

| SHOT | IN ENGLISH | RULE | TEST CLIPS WITH GOLD | SCENEGAP | SINGLE-FRAME | KEYWORDS | MISSING IN PROJECTS |
|---|---|---|---|---|---|---|---|
| **SH-01** | Storefront from the street | ≥ 3 s | 3 (36 s) | 2/5 ✓ | 3/5 ✓ | 1/2 ✓ | 3 of 8 |
| **SH-02** | Café interior, tables | ≥ 3 s | 2 (39 s) | 2/5 ✓ | 4/5 ✓ | 1/4 ✓ | 1 of 8 |
| **SH-03** | Coffee-bean close-up | ≥ 2 s | 1 (3 s) | 0/5 | 1/5 ✓ | 0/3 | 6 of 8 |
| **SH-04** | Grinding beans | ≥ 2 s | 2 (18 s) | 0/5 | 1/5 ✓ | 1/1 ✓ | 6 of 8 |
| **SH-05** | Coffee or latte art poured | ≥ 3 s | 6 (48 s) | 3/5 ✓ | 2/5 ✓ | 3/4 ✓ | 1 of 8 |
| **SH-06** | Cup of coffee on a table | ≥ 2 s | 8 (46 s) | 1/5 ✓ | 0/5 | 0/5 | 0 of 8 |
| **SH-07** | Someone drinking | ≥ 2 s | 1 (3 s) | 0/5 | 0/5 | 0/0 | 8 of 8 |
| **SH-08** | Two people, same frame | ≥ 2 s, same frame | 3 (56 s) | 4/5 ✓ | 4/5 ✓ | 0/0 | 1 of 8 |
| **SH-09** | Espresso extraction | ≥ 2 s | 2 (30 s) | 1/5 ✓ | 2/5 ✓ | 2/3 ✓ | 4 of 8 |
| **SH-10** | Bean bag, brand readable | ≥ 1 s, brand (manual) | 2 (4 s) | 0/5 | 0/5 | 0/0 | 6 of 8 |
| **SH-11** | Barista, 2 takes | ≥ 3 s, 2 takes | 5 (74 s) | 5/5 ✓ | 5/5 ✓ | 0/0 | 2 of 8 |

eval/requirements.json, eval/labels/gold.json, eval/results/test_metrics.json. ✓ = a correct clip in the top 5.

APPENDIX

# A6 · Labels: agreement, third pass and the consistency pass

## Agreement between the two AI annotators

| MEASURE | VALUE |
|---|---|
| **Clip × shot pairs agreeing (IoU ≥ 0.5 or both empty)** | 472 / 484 |
| **Frame samples (1 fps × 11 shots)** | 16,687 |
| **Both positive / A only / B only** | 808 / 46 / 97 |
| **Observed agreement / Cohen's κ** | 99.1% / 0.914 |
| **Disputed labels to third pass** | 143 |

## Consistency pass (added after labelling, before calibration)

| SHOT | AGREED POSITIVES CHECKED | REMOVED | RULE |
|---|---|---|---|
| **SH-03** | 17 | 4 | R6 |
| **SH-04** | 28 | 9 | R8 |
| **SH-09** | 41 | 0 | R2 |
| **SH-11** | 152 | 65 | R3 |

### Rules the third pass wrote (examples)

- R3 barista: commercial setting, actively preparing coffee
- R5 anything poured into the cup → SH-05, not SH-06
- R8 grinding: grounds visibly come out of the spout

> **Only removals were possible; restoring all 78 leaves SceneGap's numbers unchanged (lenient labels, A7).**

eval/labels/agreement.json, consistency.json, adjudication_rules.json.

APPENDIX

# A7 · Threshold on dev; lenient labels on test

## Dev sweep (7,282 frame × shot pairs, 432 positive)

| τ | TP | FP | FN | PRECISION | RECALL | F1 |
|---|---|---|---|---|---|---|
| **0.205** | 356 | 2,627 | 76 | 11.9% | 82.4% | 0.2085 |
| **0.210** | 351 | 2,374 | 81 | 12.9% | 81.2% | 0.2224 |
| **0.215** | 342 | 2,160 | 90 | 13.7% | 79.2% | 0.2331 |
| **0.220** | 331 | 1,992 | 101 | 14.2% | 76.6% | 0.2403 |
| **0.225** | **310** | **1,837** | **122** | **14.4%** | **71.8%** | **0.2404** |
| **0.230** | 280 | 1,671 | 152 | 14.4% | 64.8% | 0.2350 |
| **0.235** | 262 | 1,511 | 170 | 14.8% | 60.6% | 0.2376 |
| **0.240** | 239 | 1,370 | 193 | 14.9% | 55.3% | 0.2342 |
| **0.245** | 218 | 1,247 | 214 | 14.9% | 50.5% | 0.2298 |

## Test, with the 78 removed labels restored

| METHOD | HIT@5 | PRECISION@5 | MISSING CAUGHT |
|---|---|---|---|
| **Title keywords** | 6/11 (55%) | 9/22 (41%) | 33/37 (89%) |
| **Single-frame** | 8/11 (73%) | 22/55 (40%) | 0/37 (0%) |
| **SceneGap** | 7/11 (64%) | 18/55 (33%) | 0/37 (0%) |

- F1 is flat between 0.220 and 0.225; τ 0.225 chosen by the rule written before labels (ties go higher).
- Precision stays at 12–15% across the range: no single threshold separates shots well frame by frame.
- Centre crop beat padding by 0.0035 F1 (center-crop 0.2404, pad-square 0.2369).

eval/results/calibration.json, test_metrics.json (sensitivity_lenient_gold).

APPENDIX

# A8 · Missing-shot trade-off on dev (not used to rescore)

| SCORE | SETTING | MISSING CAUGHT | WRONG RESHOOT | 'FOUND' WHEN ABSENT |
|---|---|---|---|---|
| **raw cosine** | τ 0.225 | 5/89 (6%) | 0/131 (0%) | 80/89 (90%) |
| **raw cosine** | τ 0.24 | 13/89 (15%) | 7/131 (5%) | 71/89 (80%) |
| **raw cosine** | τ 0.27 | 24/89 (27%) | 14/131 (11%) | 65/89 (73%) |
| **raw cosine** | τ 0.285 | 32/89 (36%) | 22/131 (17%) | 49/89 (55%) |
| **raw cosine** | τ 0.29 | 41/89 (46%) | 40/131 (31%) | 37/89 (42%) |
| **raw cosine** | τ 0.3 | 46/89 (52%) | 62/131 (47%) | 37/89 (42%) |
| **margin over generic prompts** | 0.0 | 6/89 (7%) | 0/131 (0%) | 76/89 (85%) |
| **margin over generic prompts** | 0.0075 | 11/89 (12%) | 0/131 (0%) | 70/89 (79%) |
| **margin over generic prompts** | 0.02 | 16/89 (18%) | 7/131 (5%) | 66/89 (74%) |
| **margin over generic prompts** | 0.0475 | 29/89 (33%) | 14/131 (11%) | 51/89 (57%) |
| **margin over generic prompts** | 0.05 | 30/89 (34%) | 14/131 (11%) | 51/89 (57%) |

eval/results/explore_dev_gap_tradeoff.json: 20 dev projects × 8 clips. Margin = similarity minus the best of 6 generic prompts.

APPENDIX

# A9 · Acceptance cases: 30 of 36 pass by automated check

| STATUS | COUNT | CASES (TC-) |
|---|---|---|
| **pass** | 30 | 001, 002, 003, 004, 005, 006, 007, 008, 009, 010, 011, 012, 013, 014, 016, 019, 020, 021, 022, 023, 024, 025, 026, 027, 028, 029, 030, 031, 035, 036 |
| **not applicable** | 1 | 015 |
| **not run** | 1 | 017 |
| **partial** | 1 | 018 |
| **measured** | 1 | 032 |
| **proxy** | 1 | 033 |
| **manual check** | 1 | 034 |

> **47 automated tests; TC-035 is the clean rebuild from the public package index.**

eval/results/ACCEPTANCE.md (each case with the test that covers it), clean_env.json.

APPENDIX

# A10 · Hailo-10H model zoo: options for the device

| MODEL (HEF) | ACCURACY, FLOAT → ON CHIP | FPS BATCH 1 / 8 | ROLE |
|---|---|---|---|
| **clip_vit_b_32_image_encoder** | CIFAR-100 zero-shot 65.3 → 64.5 | 81.5 / 350 | planned: same OpenAI weights as the PC model |
| **clip_vit_b_32_text_encoder** | Retrieval@10 90.6 → 89.3 | 48.9 / 98.8 | not planned: text stays on the PC |
| **tinyclip_vit_61m_32 image** | 72.8 → 69.8 | 101 / 388 | fallback if G1 fails (new threshold) |
| **siglip2_b_32_256 image** | 74.7 → 70.4 | 64.7 / 221 | multilingual option for Chinese shot lists |
| **clip_vit_b_16 image** | 68.6 → 64.6 | 51.5 / 158 | finer detail, slower |

> **At batch 8 the vendor figure (350 FPS) is close to the RTX 4080 we measured (365 frames/s); the device's case is freeing the host, not raw speed.**

Hailo Model Zoo, Hailo-10H, PCIe Gen3 ×4 host (not USB); docs/MODEL_PLAN.md §2.

APPENDIX

# A11 · Market and price sources

| FACT | VALUE | SOURCE | VERIFIED |
|---|---|---|---|
| **US film and video editors / camera operators** | 39,400 / 32,600 (2025); 35% self-employed | US BLS OOH, accessed 2026-10-26 | yes |
| **Median pay, editors / camera operators** | $75,420 / $74,990 (May 2025) | US BLS OOH | yes |
| **Day rate, mid-level videographer (US)** | $800–1,500 | Film Lifestyle rate guide | yes (estimate) |
| **Event video, full day (Taiwan)** | NT$12,000–28,000 | PRO360 quote statistics | yes (estimate) |
| **UGen300 USB 8 GB** | $299.99 (M.2 $259.99) | CNX Software; ASUS spec page | partial |
| **Premiere Pro single app / Jumper** | $22.99/mo / from $49/mo | Adobe checkout excerpt / Jumper site | no / partial |
| **Resolve Studio / Final Cut Pro** | $295 / $299.99 one-time | Blackmagic / Apple | yes |
| **Frame.io Pro / Team; Creator Studio** | $15 / $25 per member per month; $129/yr | Frame.io / Apple | yes |
| **Axle AI FOCUS** | $2,995–28,995 per year | trade press (IBC 2026) | yes |
| **Creators worldwide** | ~50M, ~4% professional | Goldman Sachs | partial |

docs/research/BUSINESS_FACTS.md (URLs, access dates and verified flags for every row).

APPENDIX

# A12 · Engineering: 47 tests; 30 of 36 acceptance cases pass

These show the software is reliable, not that the AI is accurate. Other 6: 1 not applicable, 1 not run, 1 partial, 1 measured, 1 proxy, 1 manual check (needs a UGen300, a detector, or users).

### Media edge cases

Rotation, 29.97 fps, VFR, start offsets, corrupt and unsupported codecs, files still copying, moved files found again by hash.

### Logic

'Can't tell' stays 'can't tell'; two short clips never count as one long one; duplicates are not two takes; editing a shot voids old decisions.

### Security

127.0.0.1 only, per-run token, Origin + Host checks, authorised folders, CSV formula guard, escaped HTML, errors never expose internals.

### Offline

Core flow tested with outbound sockets blocked. Footage never leaves the computer.

### Failure

Disk full stops safely and resumes; interrupted jobs resume; originals' hashes unchanged.

### Rebuild

A fresh venv from the public index passes the test suite; Dockerfile and CI included; Apache-2.0.

eval/results/ACCEPTANCE.md lists all 36 cases with the test that covers each (A9).